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Harbor Goods: AWS cost optimization audit

Fictional sample. Harbor Goods and all data here are fictional. Each repository in this portfolio is a separate engagement with Harbor Goods, a fictional mid-size retailer. Account IDs are AWS documentation examples. The billing records, recommendations and inventories are synthetic; all savings figures estimate outcomes from them and none are measured results.

Item	Value
Client	Harbor Goods (fictional mid-size retailer)
Scope	3 AWS accounts in one AWS Organization, region us-east-1
Billing window	M1 to M3 (three monthly billing periods, oldest first)
Analysis period	M3, net unblended cost \$24,502.82
Access used	Read-only exports; no change was made to any account
Deliverable	This report, evidence/ and the scripts that produced both

1. Executive summary

AWS spending for Harbor Goods reached \$24,502.82 in M3, an increase of 9.8% (\$2,194.08) from M1. Across 16 opportunities, the audit estimates savings of **\$8,134.97 a month (\$97,619.64 a year)**, equivalent to 33.2% of the M3 bill. Completing every recommendation would bring monthly costs for the same workload to approximately \$16,367.85.

- **10 quick wins** offer \$3,889.12 in monthly savings at low effort and low risk. Most involve removing idle resources or changing a single setting; a platform engineer can complete these within a week.

- **6 planned items** offer \$4,245.85 in monthly savings. Each requires a load test, release or one-year commitment and therefore needs a change ticket and rollback plan.
- One-time expenses of \$1,012.50 cover rollback snapshots for volume deletions and S3 lifecycle transition requests. Each expense pays back within a few months.

The largest three opportunities account for these estimated monthly savings:

1. **SAV-02: Rightsize the production orders database** reduces monthly costs by \$1,693.60.
2. **SAV-03: Rightsize the staging orders database and remove its reader** reduces monthly costs by \$1,481.90.
3. **SAV-01: Rightsize over-provisioned EC2 instances** reduces monthly costs by \$840.96.

Incomplete tagging also limits cost allocation: `cost-center` appears on only 79.4% of spending in M3. Finance therefore cannot yet provide individual team bills. Section 9 describes the corrective work.

2. Scope and method

Only exports inform this audit, matching the files a client would supply through read-only access:

- **Cost and Usage Report (CUR 2.0 columns):** `data/synthetic/cur/` contains 387 line items covering 14 services, aggregated by resource and billing period.
- **AWS Compute Optimizer:** recommendations for EC2 and RDS based on a 14-day look-back.
- **AWS Trusted Advisor:** cost-optimization checks identifying idle and unattached resources.
- **Inventories:** records of EBS snapshots, AMIs, CloudWatch log groups, NAT gateways and the existing Savings Plan, plus the log bucket's S3 storage-by-age report.

The calculations follow these rules:

1. The CUR for M3 supplies every current cost. Recommendation tools identify candidate resources but supply no dollar figures.
2. Unit prices from `data/assumptions.json` determine target costs. These approximate on-demand list pricing in us-east-1; an actual engagement obtains prices from the AWS Price List API.
3. Each recommendation's monthly savings round to the nearest cent. Multiplying that rounded amount by twelve gives annual savings. Totals sum the rounded amounts, keeping tables exact.
4. Commitment sizing follows all other changes and uses only the remaining usage, preventing double counting of savings (section 7).
5. `data/assumptions.json` includes the effort and risk ratings and their rationale. An item qualifies as a quick win only when both ratings are low; all others require planned work.

The extract excludes tax, credits, refunds, AWS Support charges and Marketplace purchases, so these fall outside the audit. Architecture changes requiring redesign and pricing outside us-east-1 are also beyond scope.

3. Where the money goes

Service	M1	M2	M3	Change M1 to M3	Share of M3
Amazon EC2 (instances, EBS, NAT)	\$6,385.67	\$7,758.95	\$7,785.95	\$1,400.28 (21.9%)	31.8%
Amazon RDS	\$6,184.20	\$6,184.20	\$6,184.20	\$0.00 (0.0%)	25.2%
Amazon CloudFront	\$2,380.00	\$2,465.00	\$2,550.00	\$170.00 (7.1%)	10.4%
Amazon CloudWatch	\$1,345.00	\$1,486.00	\$1,627.00	\$282.00 (21.0%)	6.6%
Amazon Redshift	\$1,585.56	\$1,585.56	\$1,585.56	\$0.00 (0.0%)	6.5%

Service	M1	M2	M3	Change M1 to M3	Share of M3
Amazon S3	\$1,448.00	\$1,497.00	\$1,545.00	\$97.00 (6.7%)	6.3%
All other (8 services)	\$2,980.31	\$3,103.51	\$3,225.11	\$244.80 (8.2%)	13.2%
Total	\$22,308.74	\$24,080.22	\$24,502.82	\$2,194.08 (9.8%)	100.0%

Spending increased \$2,194.08 across the billing window, with the largest changes in:

- Amazon EC2 (instances, EBS, NAT), which rose \$1,400.28 as the production order-api fleet expanded from 4 to 6 instances and staging expanded from 2 to 4.
- Amazon CloudWatch, which rose \$282.00 as log groups without expiration continued accumulating data each month.
- Data transfer, which rose \$180.00 alongside monthly increases in production internet egress.

The account breakdown follows:

Account	M1	M2	M3	Change M1 to M3	Share of M3
111122223333 (production)	\$18,007.87	\$19,043.21	\$19,415.31	\$1,407.44 (7.8%)	79.2%
444455556666 (staging)	\$3,059.84	\$3,724.98	\$3,729.48	\$669.64 (21.9%)	15.2%
123456789012 (shared services)	\$1,241.03	\$1,312.03	\$1,358.03	\$117.00 (9.4%)	5.5%

4. Ranked savings

The following ranking orders recommendations by estimated monthly savings from fictional data. Recommendation IDs remain unchanged when ranks move. `evidence/savings-detail.csv` provides the calculations for individual resources.

Rank	ID	Recommendation	Monthly	Annual	One-time cost	Effort	Risk	Type
1	SAV-02	Rightsize the production orders database	\$1,693.60	\$20,323.20	none	Medium	Medium	Planned work
2	SAV-03	Rightsize the staging orders database and remove its reader	\$1,481.90	\$17,782.80	none	Low	Low	Quick win
3	SAV-01	Rightsize over-provisioned EC2 instances	\$840.96	\$10,091.52	none	Medium	Medium	Planned work
4	SAV-04	Run staging order-api instances on a working-hours schedule	\$721.92	\$8,663.04	none	Low	Low	Quick win
5	SAV-15	Buy a one-year Compute Savings Plan for the steady EC2 baseline	\$651.29	\$7,815.48	none	Low	Medium	Planned work
6	SAV-16	Buy one-year RDS Reserved Instances for production after the resize	\$635.10	\$7,621.20	none	Low	Medium	Planned work
7	SAV-11	Set CloudWatch Logs retention on groups that never expire	\$539.55	\$6,474.60	none	Low	Low	Quick win

Rank	ID	Recommendation	Monthly	Annual	One-time cost	Effort	Risk	Type
8	SAV-05	Delete unattached EBS volumes after a rollback snapshot	\$385.00	\$4,620.00	\$212.50	Low	Low	Quick win
9	SAV-13	Move application logs older than 30 days to S3 Glacier Instant Retrieval	\$378.50	\$4,542.00	\$800.00	Low	Low	Quick win
10	SAV-12	Turn off DEBUG logging in production order-api	\$360.00	\$4,320.00	none	Medium	Low	Planned work
11	SAV-10	Stop, then terminate idle untagged instances	\$141.47	\$1,697.64	none	Low	Low	Quick win
12	SAV-07	Delete EBS snapshots older than 180 days	\$122.50	\$1,470.00	none	Low	Low	Quick win
13	SAV-06	Change attached gp2 volumes to gp3	\$69.00	\$828.00	none	Low	Low	Quick win
14	SAV-14	Consolidate staging NAT gateways to one	\$64.90	\$778.80	none	Medium	Low	Planned work
15	SAV-09	Delete idle load balancers	\$34.68	\$416.16	none	Low	Low	Quick win
16	SAV-08	Release unassociated Elastic IP addresses	\$14.60	\$175.20	none	Low	Low	Quick win
Total			\$8,134.97	\$97,619.64	\$1,012.50			

5. Quick wins versus planned work

Quick wins: \$3,889.12 monthly (\$46,669.44 annually). Complete these before planned work, following rank order within a single change window. Every item either supports a direct reversal or retains a snapshot until validation confirms the change.

ID	Recommendation	Monthly	Why this effort and risk
SAV-03	Rightsize the staging orders database and remove its reader	\$1,481.90	Staging only; the writer restarts during the class change, and a short outage is acceptable.
SAV-04	Run staging order-api instances on a working-hours schedule	\$721.92	Instance Scheduler or an EventBridge schedule; staging is idle outside working hours.
SAV-11	Set CloudWatch Logs retention on groups that never expire	\$539.55	One setting per log group; export anything under a compliance hold first.
SAV-05	Delete unattached EBS volumes after a rollback snapshot	\$385.00	Snapshot each volume first, then delete; the snapshot is the rollback.
SAV-13	Move application logs older than 30 days to S3 Glacier Instant Retrieval	\$378.50	One lifecycle rule; reads of older logs are rare and remain immediate in Glacier Instant Retrieval.
SAV-10	Stop, then terminate idle untagged instances	\$141.47	No owner tag and about 1% CPU; stop, wait one week for a claim, then terminate.

ID	Recommendation	Monthly	Why this effort and risk
SAV-07	Delete EBS snapshots older than 180 days	\$122.50	Only snapshots older than the threshold, not used by an AMI and without a retention tag.
SAV-06	Change attached gp2 volumes to gp3	\$69.00	Elastic Volumes change the type online, with no detach and no downtime.
SAV-09	Delete idle load balancers	\$34.68	No requests in 14 days; delete after checking DNS.
SAV-08	Release unassociated Elastic IP addresses	\$14.60	Release after confirming no DNS record or allow list points at the address.

Planned work: \$4,245.85 monthly (\$50,950.20 annually). Follow this order:

1. Complete rightsizing (SAV-02, SAV-01) before purchasing commitments. Each resize requires a load test and maintenance window, with a return to the previous instance class available for rollback.
2. Purchase commitments (SAV-15, SAV-16) against the resulting usage. An earlier purchase would commit to the oversized fleet for one year.
3. Include the logging update (SAV-12) in the next order-api release. Implement staging NAT gateway consolidation (SAV-14) through the staging network code.

ID	Recommendation	Monthly	Why this effort and risk
SAV-02	Rightsize the production orders database	\$1,693.60	Resize the reader, fail over to it, then resize the old writer; schedule it in a maintenance window after a load test.
SAV-01	Rightsize over-provisioned EC2 instances	\$840.96	Instance type change needs a rolling replacement and a load test on memory headroom.
SAV-15	Buy a one-year Compute Savings Plan for the steady EC2 baseline	\$651.29	A one-year commitment; buy it after rightsizing so it does not lock in oversized usage.
SAV-16	Buy one-year RDS Reserved Instances for production after the resize	\$635.10	A one-year commitment; buy it after the orders-db resize.
SAV-12	Turn off DEBUG logging in production order-api	\$360.00	Needs an application config change and a release of order-api.
SAV-14	Consolidate staging NAT gateways to one	\$64.90	Route table changes in staging; an AZ outage would cut staging egress, which is acceptable.

6. Idle, unattached and rightsizing findings

6.1 Idle and unattached resources

These resources come from Trusted Advisor findings. Snapshot selection applies an additional rule: snapshots must exceed 180 days in age, have no AMI reference and carry no **retention** tag. This excludes 1 legal-hold snapshot and 2 snapshots backing AMIs.

ID	Account	Resource	Detail	Current/mo	Target/mo	Saving/mo
SAV-05	111122223333	vol-0a9000000000000e1	gp2, 1,000 GiB, left over from the app fleet rebuild	\$100.00	\$0.00	\$100.00
SAV-05	111122223333	vol-0a9000000000000e2	gp2, 500 GiB, detached test restore	\$50.00	\$0.00	\$50.00
SAV-05	444455556666	vol-0b9000000000000e1	gp2, 500 GiB, old staging database volume	\$50.00	\$0.00	\$50.00
SAV-05	123456789012	vol-0c9000000000000e1	gp3, 2,000 GiB, retired build cache	\$160.00	\$0.00	\$160.00
SAV-05	123456789012	vol-0c9000000000000e2	gp2, 250 GiB, detached from a terminated dev box	\$25.00	\$0.00	\$25.00
SAV-07	111122223333	snap-0a1000000000000004	900 GiB billed, 420 days old	\$45.00	\$0.00	\$45.00
SAV-07	111122223333	snap-0a1000000000000006	650 GiB billed, 610 days old	\$32.50	\$0.00	\$32.50
SAV-07	444455556666	snap-0b1000000000000001	400 GiB billed, 300 days old	\$20.00	\$0.00	\$20.00
SAV-07	123456789012	snap-0c1000000000000001	500 GiB billed, 250 days old	\$25.00	\$0.00	\$25.00
SAV-08	111122223333	eipalloc-0a1000000000000001	public IP 203.0.113.10	\$3.65	\$0.00	\$3.65
SAV-08	111122223333	eipalloc-0a1000000000000002	public IP 203.0.113.11	\$3.65	\$0.00	\$3.65
SAV-08	444455556666	eipalloc-0b1000000000000001	public IP 203.0.113.20	\$3.65	\$0.00	\$3.65
SAV-08	123456789012	eipalloc-0c1000000000000001	public IP 203.0.113.30	\$3.65	\$0.00	\$3.65
SAV-09	111122223333	legacy-promo-alb	application, 0 requests in 14 days	\$16.43	\$0.00	\$16.43
SAV-09	444455556666	stg-legacy-clb	classic, 0 requests in 14 days	\$18.25	\$0.00	\$18.25
SAV-10	123456789012	i-0c2000000000000004	dev-04 (t3.large), 1.2% average CPU	\$60.74	\$0.00	\$60.74
SAV-10	123456789012	vol-0c2000000000000004	root volume of dev-04	\$10.00	\$0.00	\$10.00
SAV-10	123456789012	i-0c2000000000000005	dev-05 (t3.large), 1.2% average CPU	\$60.74	\$0.00	\$60.74
SAV-10	123456789012	vol-0c2000000000000005	root volume of dev-05	\$10.00	\$0.00	\$10.00

Create a snapshot of every one of the 5 unattached volumes before deletion, then retain those snapshots for one month. The one-time cost is \$212.50, an upper bound assuming full volume size, with payback in 0.6 months.

6.2 Rightsizing and scheduling

Compute Optimizer classified both orders databases and the order-api instances as over-provisioned. Because staging order-api instances handle working-day activity, the recommendation schedules 260 running hours per month while retaining their current instance type.

ID	Account	Resource	Detail	Current/mo	Target/mo	Saving/mo
SAV-01	111122223333	i-0a2000000000000001	app-01: m5.2xlarge to m5.xlarge (max CPU 17%, max memory 34%)	\$280.32	\$140.16	\$140.16
SAV-01	111122223333	i-0a2000000000000002	app-02: m5.2xlarge to m5.xlarge (max CPU 17%, max memory 34%)	\$280.32	\$140.16	\$140.16

ID	Account	Resource	Detail	Current/mo	Target/mo	Saving/mo
SAV-01	111122223333	i-0a2000000000000003	app-03: m5.2xlarge to m5.xlarge (max CPU 17%, max memory 34%)	\$280.32	\$140.16	\$140.16
SAV-01	111122223333	i-0a2000000000000004	app-04: m5.2xlarge to m5.xlarge (max CPU 17%, max memory 34%)	\$280.32	\$140.16	\$140.16
SAV-01	111122223333	i-0a2000000000000005	app-05: m5.2xlarge to m5.xlarge (max CPU 17%, max memory 34%)	\$280.32	\$140.16	\$140.16
SAV-01	111122223333	i-0a2000000000000006	app-06: m5.2xlarge to m5.xlarge (max CPU 17%, max memory 34%)	\$280.32	\$140.16	\$140.16
SAV-02	111122223333	orders-db	db.r5.4xlarge writer and reader to db.r5.2xlarge writer and reader	\$3,387.20	\$1,693.60	\$1,693.60
SAV-03	444455556666	orders-db-stg	db.r5.2xlarge writer and reader to db.r5.large writer only	\$1,693.60	\$211.70	\$1,481.90
SAV-04	444455556666	i-0b1000000000000001	stg-app-01 (m5.2xlarge)	\$280.32	\$99.84	\$180.48
SAV-04	444455556666	i-0b1000000000000002	stg-app-02 (m5.2xlarge)	\$280.32	\$99.84	\$180.48
SAV-04	444455556666	i-0b1000000000000003	stg-app-03 (m5.2xlarge)	\$280.32	\$99.84	\$180.48
SAV-04	444455556666	i-0b1000000000000004	stg-app-04 (m5.2xlarge)	\$280.32	\$99.84	\$180.48

The remaining fleets have approximately appropriate capacity; the table shows the first instance in each. Load tests drive staging CPU to 65%, supporting its current instance size.

Resource	Current type	Compute Optimizer finding	Max CPU	Max memory
report-01	m5.xlarge	Optimized	41%	58%
batch-01	c5.4xlarge	Optimized	88%	61%
stg-app-01	m5.2xlarge	Optimized	65%	48%
ci-runner-01	c5.xlarge	Optimized	92%	40%

7. Savings Plans and Reserved Instances

CUR line types establish the following coverage for M3:

Measure	Value
Savings Plans eligible on-demand spend (M3)	\$6,752.37
Covered by the existing Compute Savings Plan	\$560.64
Savings Plans coverage	8.3%
Existing plan utilization	100.0%
Observed Savings Plan discount (1 - fee / covered)	27.1%
RDS instance-hour spend (M3)	\$5,504.20

Measure	Value
RDS Reserved Instance coverage	0.0%

The existing Compute Savings Plan is fully utilized and applies exclusively to the four **m5.xlarge** instances of the reporting worker fleet, giving Harbor Goods 8.3% coverage. The storefront web tier runs on ECS Fargate; its tasks are eligible but run on demand. The next plan's sizing uses the observed 27.1% discount because it reflects the instance families Harbor Goods runs.

For the proposed Compute Savings Plan (SAV-15), sizing follows these criteria:

- Include only instances that ran continuously throughout all three billing periods, using their capacity after rightsizing. Exclude scheduled, idle and untagged instances from this baseline.
- Set the commitment at 80% of baseline usage, leaving room for additional rightsizing or migration to Graviton or containers.
- Round down the hourly commitment to a valid purchase increment.
- Leave the storefront's Fargate spend out of the baseline. A Compute Savings Plan also applies to Fargate, so that usage absorbs any commitment that further EC2 rightsizing frees.

Step	Value
Steady post-change baseline (14 instances, on-demand per hour)	\$4.1192
Target covered on-demand per hour	\$3.2954
Commitment per hour (rounded down)	\$2.4020
On-demand equivalent covered per hour	\$3.2942
Commitment per month	\$1,753.46
Estimated saving per month	\$651.29

EC2 changes reduce eligible monthly on-demand spending to \$5,068.02. Together, the existing and proposed plans would cover 58.5% of that spending.

The following instances form the baseline:

Account	Instance	Sized as	On-demand per hour
111122223333	i-0a2000000000000001	m5.xlarge	\$0.1920
111122223333	i-0a2000000000000002	m5.xlarge	\$0.1920
111122223333	i-0a2000000000000003	m5.xlarge	\$0.1920
111122223333	i-0a2000000000000004	m5.xlarge	\$0.1920
111122223333	i-0a3000000000000001	c5.4xlarge	\$0.6800
111122223333	i-0a3000000000000002	c5.4xlarge	\$0.6800
111122223333	i-0a3000000000000003	c5.4xlarge	\$0.6800
111122223333	i-0a3000000000000004	c5.4xlarge	\$0.6800
111122223333	i-0a4000000000000001	t3.medium	\$0.0416
123456789012	i-0c1000000000000001	c5.xlarge	\$0.1700

Account	Instance	Sized as	On-demand per hour
123456789012	i-0c1000000000000002	c5.xlarge	\$0.1700
123456789012	i-0c2000000000000001	t3.large	\$0.0832
123456789012	i-0c2000000000000002	t3.large	\$0.0832
123456789012	i-0c2000000000000003	t3.large	\$0.0832

RDS Reserved Instances (SAV-16). Current RDS coverage stands at 0.0%. Once orders-db has been resized, reserve both production databases under one-year terms with no upfront payment. A 30% discount supplies the planning estimate; verify the actual rate through the Price List API before purchase. Aurora PostgreSQL Reserved Instances support size flexibility within an instance family. Purchasing after resizing still aligns the commitment with the running resources.

ID	Account	Resource	Detail	Current/mo	Target/mo	Saving/mo
SAV-16	111122223333	orders-db	db.r5.2xlarge writer and reader	\$1,693.60	\$1,185.52	\$508.08
SAV-16	111122223333	reporting-db	db.r5.xlarge writer only	\$423.40	\$296.38	\$127.02

This audit models both commitments without purchasing either. Harbor Goods retains responsibility for the decision and purchase.

8. Storage, logs and network

ID	Account	Resource	Detail	Current/mo	Target/mo	Saving/mo
SAV-11	111122223333	/harbor-goods/order-api	application, never expires to 30 days, 9,000 GB stored to 360 GB	\$270.00	\$10.80	\$259.20
SAV-11	111122223333	/harbor-goods/storefront	application, never expires to 30 days, 3,600 GB stored to 90 GB	\$108.00	\$2.70	\$105.30
SAV-11	111122223333	/harbor-goods/vpc-flow	network, never expires to 180 days, 4,800 GB stored to 720 GB	\$144.00	\$21.60	\$122.40
SAV-11	444455556666	/harbor-goods/order-api-stg	application, never expires to 30 days, 1,800 GB stored to 45 GB	\$54.00	\$1.35	\$52.65
SAV-12	111122223333	/harbor-goods/order-api	1,200 GB a month ingested, 60% assumed DEBUG	\$600.00	\$240.00	\$360.00
SAV-13	123456789012	harbor-goods-app-logs	20,000 GB in 40,000,000 objects older than 30 days	\$460.00	\$80.00	\$380.00
SAV-13			Less retrieval fees for 50 GB of reads a month			-\$1.50
SAV-14	444455556666	nat-0b1000000000000002	stg-nat-b hourly charge	\$32.85	\$0.00	\$32.85
SAV-14	444455556666	nat-0b1000000000000003	stg-nat-c hourly charge	\$32.85	\$0.00	\$32.85
SAV-14			Less cross-AZ transfer for 40 GB moved to the remaining gateway			-\$0.80

- **CloudWatch Logs retention (SAV-11).** No expiration applies to 4 log groups. The calculation assumes stored volume stabilizes at monthly ingest multiplied by 0.30 and retention months, with the ratio representing compression. Set application log retention to 30 days and VPC flow log retention to 180 days, first exporting any data subject to a compliance hold to S3.
- **DEBUG logging (SAV-12).** The calculation assumes DEBUG entries make up about 60% of order-api log volume. Validate that share through a CloudWatch Logs Insights query before release. Excluding the storage impact keeps the savings estimate conservative.
- **S3 lifecycle (SAV-13).** Transition log objects after 30 days to S3 Glacier Instant Retrieval, preserving millisecond read access. Estimated savings deduct the cost of 50 GB in monthly reads. A one-time transition-request expense of \$800.00 pays back over 2.1 months.
- **gp2 to gp3 (SAV-06).** Convert 18 attached gp2 volumes online. All are at or below 1,000 GiB, so the included gp3 capacity of 3,000 IOPS meets or exceeds their gp2 baseline. Volumes larger than 170 GiB receive additional provisioned throughput to preserve gp2's 250 MiB/s; the calculation includes that expense. Exclude unattached volumes and volumes belonging to idle instances, since those are scheduled for deletion.
- **Staging NAT gateways (SAV-14).** Each of the 3 staging gateways handles no more than 20 GB monthly. A single gateway can handle the workload; estimated savings account for the additional cross-AZ transfer cost.

9. Tagging and cost allocation

Required tag	Share of M3 cost tagged	Untagged cost
application	89.8%	\$2,494.25
environment	88.5%	\$2,809.61
cost-center	79.4%	\$5,044.25

Spending without tags in M3 has three components. Resources account for \$1,185.45, which resource tagging can address. Account-level data transfer accounts for \$900.00, while Savings Plan fees account for \$408.80. Those last two components cannot carry resource tags and require allocation rules.

The highest-cost resources without tags appear below:

Account	Service	Resource	M3 cost
444455556666	Amazon EC2 (instances, EBS, NAT)	i-0b2000000000000001	\$248.20
123456789012	Amazon EC2 (instances, EBS, NAT)	vol-0c90000000000000e1	\$160.00
444455556666	Amazon CloudWatch	/harbor-goods/order-api-stg	\$129.00
111122223333	Amazon EC2 (instances, EBS, NAT)	vol-0a90000000000000e1	\$100.00
123456789012	Amazon EC2 (instances, EBS, NAT)	i-0c2000000000000004	\$60.74
123456789012	Amazon EC2 (instances, EBS, NAT)	i-0c2000000000000005	\$60.74
111122223333	Amazon EC2 (instances, EBS, NAT)	vol-0a90000000000000e2	\$50.00
444455556666	Amazon EC2 (instances, EBS, NAT)	vol-0b2000000000000001	\$50.00

The recommended allocation controls are:

1. **Require three tags:** **application**, **environment** and **cost-center**. Publish permitted values in a single location, and retain **Name** for human identification rather than cost allocation.

2. **Apply creation-time controls:** use an AWS Organizations tag policy to govern allowed values. Separately, require tags in Terraform modules. An AWS Config `required-tags` rule reports drift.
3. **Enable cost allocation for all three keys** in the management account, making them available in Cost Explorer and the CUR.
4. **Use AWS Cost Categories for costs without resource tags:** apportion account-level data transfer according to each application's share of CloudFront and ELB usage. Allocate the Savings Plan fee according to each cost center's covered usage.
5. **Assign responsibility for unowned resources:** give every resource above an owner or deletion date during the first quick-win window. The list already includes the idle instances (SAV-10).
6. **Produce monthly showback reports** for each cost center and target 95% tagged spend.

10. Assumptions

`data/assumptions.json` holds all assumptions. Updating a value and running `make evidence` recalculates its impact throughout the report.

Assumption	Value	Used by
Hours in a month	730	all
Staging schedule	12 h x 5 days = 260 h/month	SAV-04
Snapshot age threshold	180 days	SAV-07
Rollback snapshot kept for	1 month	SAV-05
Compute Savings Plan coverage target	80% of the steady baseline	SAV-15
Compute Savings Plan discount	27.1% (observed)	SAV-15
RDS Reserved Instance discount	30% (planning figure)	SAV-16
CloudWatch Logs stored-to-ingested ratio	0.30	SAV-11
Retention: application / network logs	30 days / 180 days	SAV-11
DEBUG share of order-api log volume	60%	SAV-12
S3 transition age	30 days	SAV-13
Expected Glacier Instant Retrieval reads	50 GB/month	SAV-13
Staging NAT gateways kept	1	SAV-14

That file also contains unit prices approximating us-east-1 on-demand list pricing. Because Savings Plan and Reserved Instance rates vary, verify them through the AWS Price List API before making any purchase.

11. Reproduce this report

```
make verify    # tests, evidence and report freshness, PDF freshness, lint
make evidence  # regenerate evidence/ and this report from data/
```

The referenced evidence resides in `evidence/`, including `spend-by-service.csv`, `spend-by-account.csv`, `savings-ranked.csv`, `savings-detail.csv`, `tag-coverage.csv`, `commitments.json` and `facts.json`. Checks in `tests/` independently recalculate every figure from `data/` and fail when report contents diverge from those inputs.

Fictional sample: This report uses an invented company, Harbor Goods, and entirely synthetic data. All savings figures are estimates rather than measured outcomes.